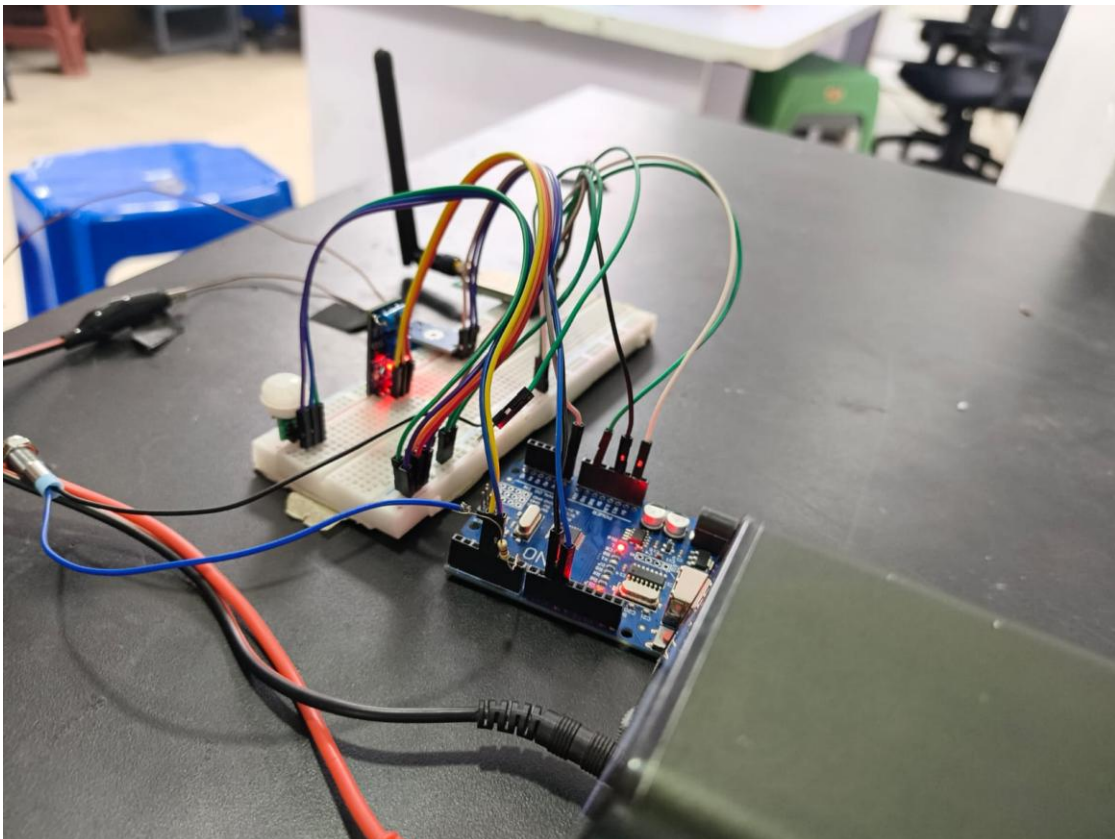


ANTI-THEFT VEHICLE BATTERY DETECTION SYSTEM

Complete Project Documentation

Components Used:

Arduino Uno | LoRa E32-433T20D | HC-SR505 PIR Sensor
SW-420 Vibration Sensor | Voltage Sensor | Active Buzzer



1. Project Overview

The Anti-Theft Vehicle Battery Detection System is a wireless alarm system designed to protect vehicle batteries from theft and unauthorized removal. It uses two Arduino Uno microcontrollers communicating over LoRa (Long Range) radio to detect suspicious activity and alert the vehicle owner remotely in real time.

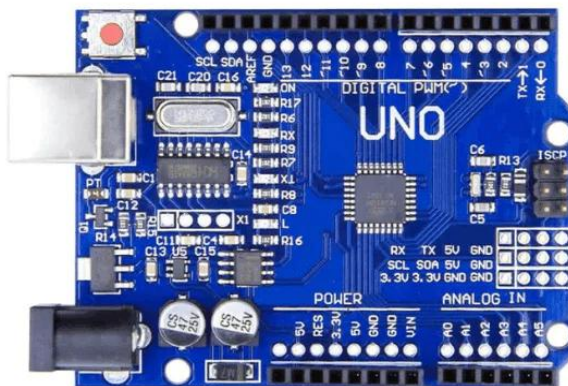
The system monitors the battery area using three sensors — the HC-SR505 PIR motion sensor, the SW-420 vibration sensor, and a voltage sensor. The buzzer at the battery location (transmitter) scares the thief on-site, while the buzzer at the owner's location (receiver) gives a remote alert simultaneously.

Key Feature: The alarm only triggers when BOTH the PIR sensor AND the vibration sensor detect activity together — preventing false alarms from motion alone or vibration alone.

2. Components Used

2.1 Arduino Uno (x2)

Why Used: The Arduino Uno is the brain of the entire system. One Arduino is placed at the transmitter side (battery location) to read all sensors and send LoRa alerts. The second Arduino is at the receiver side (with the owner) to receive alerts and control the buzzer. It processes all sensor inputs and makes all decisions.



2.2 LoRa E32-433T20D Module (x2) — EBYTE

Why Used: LoRa (Long Range) radio allows the transmitter and receiver to communicate wirelessly up to 3km without WiFi, Bluetooth, or a SIM card. One module is on each Arduino. They exchange small text messages — ALERT, BATTERY_LOST, and BATTERY_OK — completely offline.



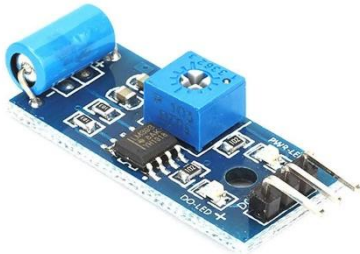
2.3 HC-SR505 PIR Motion Sensor

Why Used: The HC-SR505 detects human body infrared heat radiation. When a person approaches the vehicle battery, the sensor output goes HIGH and signals the Arduino. This is the first layer of detection — it detects that someone is near the battery. Alone it does NOT trigger the alarm; it must occur together with the vibration sensor.



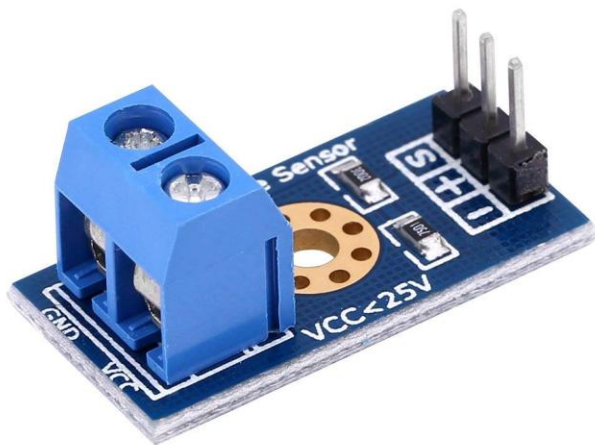
2.4 SW-420 Vibration Sensor Module

Why Used: The SW-420 vibration sensor detects physical shaking, tapping, or movement of the battery. When a thief tries to loosen bolts, lift, or physically disturb the battery, the SW-420 triggers and outputs HIGH. Combined with the PIR sensor detecting nearby presence, this confirms actual tampering rather than a false alarm.



2.5 Voltage Sensor Module (Max 25V)

Why Used: This is the most critical sensor. It continuously monitors the EV battery voltage. When the battery is disconnected or stolen, the voltage instantly drops to near 0V. The Arduino detects this and immediately triggers a continuous non-stop alarm on both sides. When the battery is reconnected, voltage rises and the alarm stops automatically.



3. Circuit Connections

3.1 Transmitter Side — At Battery Location

Component	Component Pin	Arduino Uno Pin
HC-SR505 PIR Sensor	VCC	5V
HC-SR505 PIR Sensor	GND	GND
HC-SR505 PIR Sensor	OUT	D4
SW-420 Vibration Sensor	VCC	5V
SW-420 Vibration Sensor	GND	GND
SW-420 Vibration Sensor	DO (Digital Out)	D3
Voltage Sensor	VCC	5V
Voltage Sensor	GND	GND
Voltage Sensor	S (Signal)	A0
Voltage Sensor	+ Input Terminal	Vehicle Battery (+)
Voltage Sensor	- Input Terminal	Vehicle Battery (-)
LoRa E32	VCC	3.3V (NOT 5V!)
LoRa E32	GND	GND
LoRa E32	TXD	D10 (SoftwareSerial RX)
LoRa E32	RXD	D11 (SoftwareSerial TX)
LoRa E32	M0	GND
LoRa E32	M1	GND
Buzzer	+ Red wire	D7
Buzzer	- Black wire	GND
9V Battery (separate)	+	Arduino VIN
9V Battery (separate)	-	Arduino GND

3.2 Receiver Side — With Vehicle Owner

Component	Component Pin	Arduino Uno Pin
LoRa E32	VCC	3.3V (NOT 5V!)
LoRa E32	GND	GND

Component	Component Pin	Arduino Uno Pin
LoRa E32	TXD	D10 (SoftwareSerial RX)
LoRa E32	RXD	D11 (SoftwareSerial TX)
LoRa E32	M0	GND
LoRa E32	M1	GND
Buzzer	+ Red wire	D8
Buzzer	- Black wire	GND

4. How the Project Works

4.1 System Architecture

The system has two independent Arduino units communicating over LoRa radio:

- Transmitter Unit — permanently placed at the vehicle battery. Powered by its own 9V battery so it always stays on, even when the vehicle battery is removed.
- Receiver Unit — kept with the vehicle owner. Powered via USB or 9V battery.
- LoRa Radio Link — 433 MHz wireless communication, up to 3 km range, no internet required.

4.2 Normal State — No Theft

- HC-SR505 PIR output = LOW — no body heat detected
- SW-420 vibration output = LOW — no vibration detected
- Voltage sensor reads normal battery voltage (above threshold)
- Both Arduinos are silently monitoring all sensors
- Both buzzers are completely silent

4.3 Theft Attempt — Motion AND Vibration Together

The alarm ONLY triggers when BOTH sensors activate at the same time:

- Step 1: Thief approaches — HC-SR505 PIR detects body heat → output = HIGH
- Step 2: Thief touches battery — SW-420 detects vibration → output = HIGH
- Step 3: Arduino checks: pirVal == HIGH AND vibrVal == HIGH (both together)
- Step 4: Transmitter buzzer beeps 5 times on-site — scares the thief
- Step 5: LoRa sends ALERT message to receiver
- Step 6: Receiver buzzer also beeps 5 times — alerts the owner
- Step 7: Both buzzers go silent, system resumes monitoring

If only PIR triggers (someone walks past) → No alarm. If only vibration triggers (wind shakes the vehicle) → No alarm. Both MUST happen together.

4.4 Battery Disconnected — Continuous Alarm

- Step 1: Thief disconnects or removes the vehicle battery
- Step 2: Voltage drops from normal (e.g. 12V) to near 0V instantly
- Step 3: Voltage sensor signal to A0 drops below threshold (10V)
- Step 4: batteryLost flag set to TRUE
- Step 5: Transmitter buzzer sounds CONTINUOUSLY non-stop on-site
- Step 6: LoRa sends BATTERY_LOST message to receiver
- Step 7: Receiver buzzer also sounds CONTINUOUSLY non-stop
- Step 8: Both buzzers never stop until battery is reconnected

4.5 Battery Reconnected — Alarm Stops Automatically

- Step 1: Battery is reconnected — voltage rises back above threshold
- Step 2: Arduino detects voltage $\geq 10V$ AND batteryLost == true
- Step 3: batteryLost flag reset to FALSE
- Step 4: Transmitter buzzer stops immediately
- Step 5: LoRa sends BATTERY_OK message to receiver
- Step 6: Receiver Arduino receives BATTERY_OK
- Step 7: Receiver buzzer stops immediately
- Step 8: System returns to normal silent monitoring

5. Transmitter Arduino Code

Upload this code to the Arduino Uno placed at the vehicle battery location.

Code Fix: SW-420 vibration sensor outputs HIGH when vibration is detected. The condition checks `pirVal == HIGH AND vibrVal == HIGH` together — both must be true at the same time to trigger the alarm.

```
#include <SoftwareSerial.h>

SoftwareSerial LoRa(10, 11); // RX=10, TX=11

#define PIR_PIN      4      // HC-SR505 PIR sensor
#define VIBR_PIN     3      // SW-420 Vibration sensor
#define VOLT_PIN     A0     // Voltage sensor
#define BUZZER_PIN   7      // Buzzer

#define VOLT_RATIO    5.0
#define BATTERY_THRESHOLD 10.0

bool batteryLost = false;

void beepBuzzer(int times) {
    for (int i = 0; i < times; i++) {
        digitalWrite(BUZZER_PIN, HIGH);
        delay(300);
        digitalWrite(BUZZER_PIN, LOW);
        delay(300);
    }
}

void setup() {
    Serial.begin(9600);
    LoRa.begin(9600);
    pinMode(PIR_PIN, INPUT);
    pinMode(VIBR_PIN, INPUT);
    pinMode(BUZZER_PIN, OUTPUT);
    digitalWrite(BUZZER_PIN, LOW);
    Serial.println("Transmitter Ready");
}

void loop() {
    int pirVal = digitalRead(PIR_PIN);
    int vibrVal = digitalRead(VIBR_PIN);
    int rawVolt = analogRead(VOLT_PIN);
    float voltage = (rawVolt / 1023.0) * 5.0 * VOLT_RATIO;

    Serial.print("PIR: "); Serial.print(pirVal);
    Serial.print(" VIBR: "); Serial.print(vibrVal);
    Serial.print(" VOLT: "); Serial.println(voltage);

    // Battery disconnected - buzz continuously
    if (voltage < BATTERY_THRESHOLD) {
        batteryLost = true;
        digitalWrite(BUZZER_PIN, HIGH);
    }
}
```

```
    LoRa.println("BATTERY_LOST");
    delay(500);
}

// Battery reconnected - stop buzzer
if (voltage >= BATTERY_THRESHOLD && batteryLost) {
    batteryLost = false;
    digitalWrite(BUZZER_PIN, LOW);
    LoRa.println("BATTERY_OK");
    Serial.println("Battery reconnected - buzzer stopped!");
}

// BOTH PIR (HC-SR505) AND Vibration (SW-420) must trigger TOGETHER
// SW-420 outputs HIGH when vibration detected
// HC-SR505 outputs HIGH when motion detected
if (!batteryLost && pirVal == HIGH && vibrVal == HIGH) {
    Serial.println("ALERT - Motion + Vibration detected TOGETHER!");
    beepBuzzer(5);
    LoRa.println("ALERT");
    delay(2000); // Cooldown to prevent repeated triggers
}

delay(300);
}
```

6. Receiver Arduino Code

Upload this code to the Arduino Uno kept with the vehicle owner.

```
#include <SoftwareSerial.h>

SoftwareSerial LoRa(10, 11); // RX=10, TX=11

#define BUZZER_PIN 8

bool batteryLost = false;

void beepBuzzer(int times) {
  for (int i = 0; i < times; i++) {
    digitalWrite(BUZZER_PIN, HIGH);
    delay(300);
    digitalWrite(BUZZER_PIN, LOW);
    delay(300);
  }
}

void setup() {
  Serial.begin(9600);
  LoRa.begin(9600);
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);
  Serial.println("Receiver Ready");
}

void loop() {

  // Keep buzzer on continuously when battery is lost
  if (batteryLost) {
    digitalWrite(BUZZER_PIN, HIGH);
  }

  if (LoRa.available()) {
    String msg = LoRa.readStringUntil('\n');
    msg.trim();

    Serial.print("Received: ");
    Serial.println(msg);

    // Both HC-SR505 + SW-420 triggered on transmitter side
    if (msg == "ALERT") {
      Serial.println("Alert! Beeping 5 times.");
      beepBuzzer(5); // Beep 5 times then silent
    }

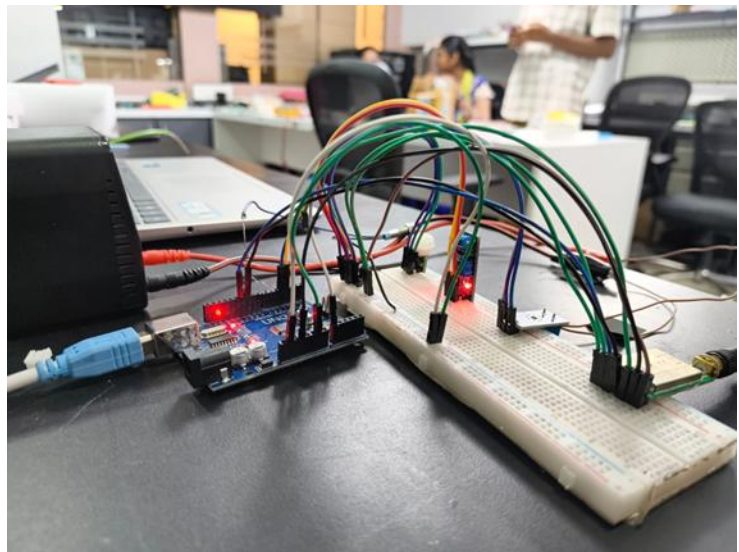
    // Battery disconnected
    if (msg == "BATTERY_LOST") {
      Serial.println("Battery removed! Continuous alarm!");
      batteryLost = true;
      digitalWrite(BUZZER_PIN, HIGH);
    }

    // Battery reconnected - stop buzzer
```

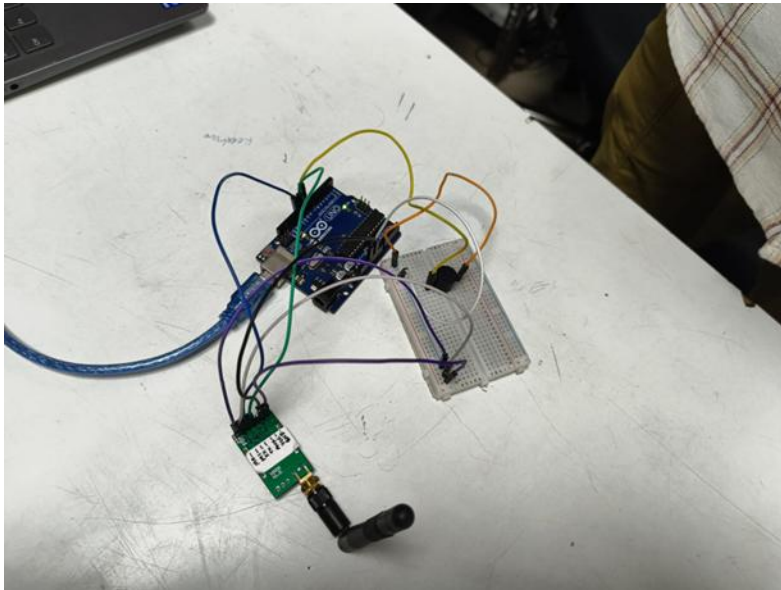
```
    if (msg == "BATTERY_OK") {  
        Serial.println("Battery reconnected. Alarm stopped.");  
        batteryLost = false;  
        digitalWrite(BUZZER_PIN, LOW);  
    }  
}  
  
delay(100);  
}
```

7.MAKING PROCEDURE

- The required hardware components such as Arduino Uno, LoRa E32-433T20D modules, HC-SR505 PIR sensor, SW-420 vibration sensor, voltage sensor module, active buzzer, breadboard, jumper wires, and power supply were collected. Connect all components as per the circuit connection tables in Section 3
- The transmitter circuit was assembled by connecting the PIR sensor, vibration sensor, voltage sensor, buzzer, and LoRa module to the Arduino Uno according to the circuit diagram. Connect a 9V test battery to the voltage sensor input terminals
- A receiver unit was built using another Arduino Uno, LoRa module, and buzzer to receive wireless alerts from the transmitter section. Open Serial Monitor at 9600 baud on both Arduinos
- The transmitter and receiver LoRa modules were configured and interfaced through serial communication to enable long-range data transmission. At the same time tap or shake the SW-420 vibration sensor module
- Arduino programs for both transmitter and receiver units were developed and uploaded using the Arduino IDE. Serial Monitor should display: ALERT - Motion + Vibration detected TOGETHER!
- The voltage sensor was calibrated to monitor the battery voltage continuously. A threshold voltage of 10V was set to identify battery disconnection or theft conditions. Disconnect the 9V test battery from the voltage sensor input terminals



- The PIR sensor and vibration sensor were tested individually and then integrated so that an alert is generated only when both motion and vibration are detected simultaneously, reducing false alarms. Serial Monitor should show: Battery removed! Continuous alarm!
- The complete system was tested by simulating vehicle battery theft conditions. When the battery voltage dropped below 10V, the transmitter generated an alert message and activated the buzzer. Both buzzers should stop automatically
- The alert information was transmitted through the LoRa module and successfully received by the receiver unit, where a warning indication was generated
- .
- The serial monitor was used to verify sensor readings, battery voltage values, and LoRa communication messages during testing.



Conclusion

The Anti-Theft Vehicle Battery Detection System was successfully designed and implemented using Arduino Uno, LoRa communication modules, PIR and vibration sensors, a voltage sensor, and buzzers. The system effectively detects unauthorized battery tampering by monitoring motion, vibration, and battery voltage conditions. When suspicious activity is detected or the battery voltage drops below the preset threshold of 10V, an alert is generated and transmitted wirelessly through LoRa communication. The receiver unit successfully receives the alert and activates the alarm, enabling remote monitoring of the vehicle battery. The project demonstrates a reliable, low-cost, and long-range security solution for protecting vehicle batteries from theft and unauthorized removal.

